

Lec 0
Syllabus
2025

algorithms
Lec-1
1.57, 2+

algorithm
PDF
C++

Lec_1

Introduction to Algorithms

Tee

Definition: An algorithm is a procedure used for solving a particular problem

Or

Set of finite Steps or rules to solve a Given Problem

~~programming~~

Example of Algorithm

Tea?

Program

C++

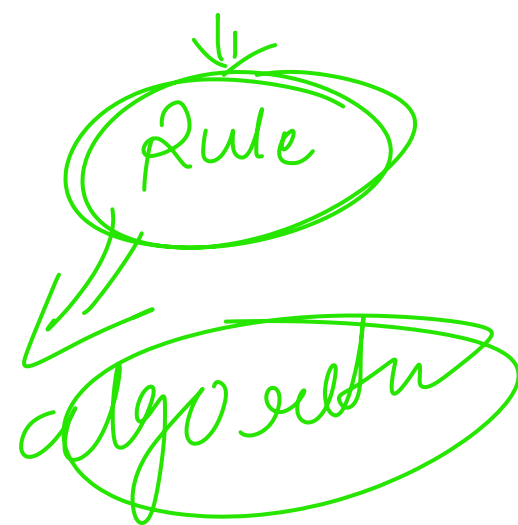
- Addition of two numbers n_1, n_2

Step 1:- Take two numbers from user

$\Rightarrow n_3 = n_1 + n_2$

$\Rightarrow \text{print}(n_3)$

$\text{return}(n_3)$



Need: - Rule, Algorithm, Program, C++, data, C++ software

Properties of an Algorithm :-

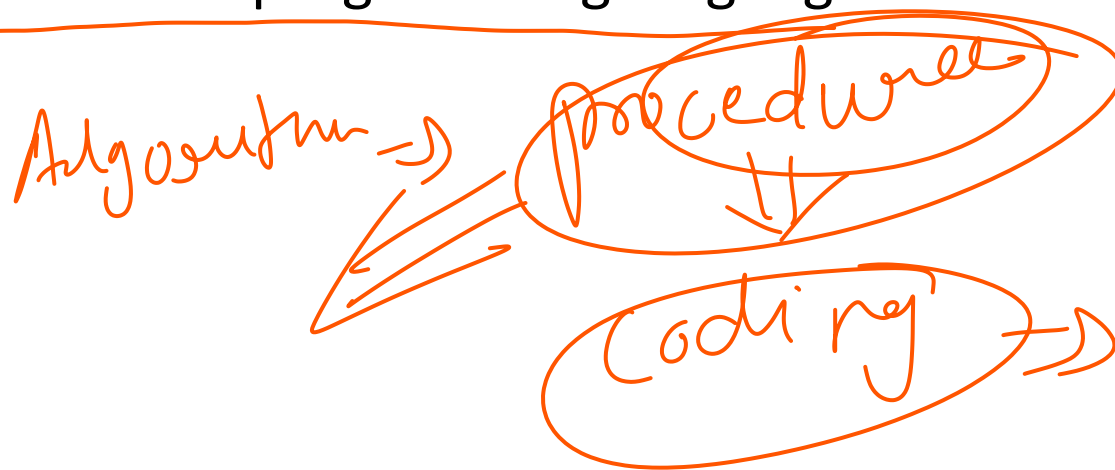
- ✓ An Algorithm Should produce at least one output
- ✓ It should have terminating point after some steps
- Every step of an algorithm should be effective
- It is independent of programming language

Result

1 ✓
2 ✓
3 ✓
4 ✓
⋮
n

~~∞~~ ✗

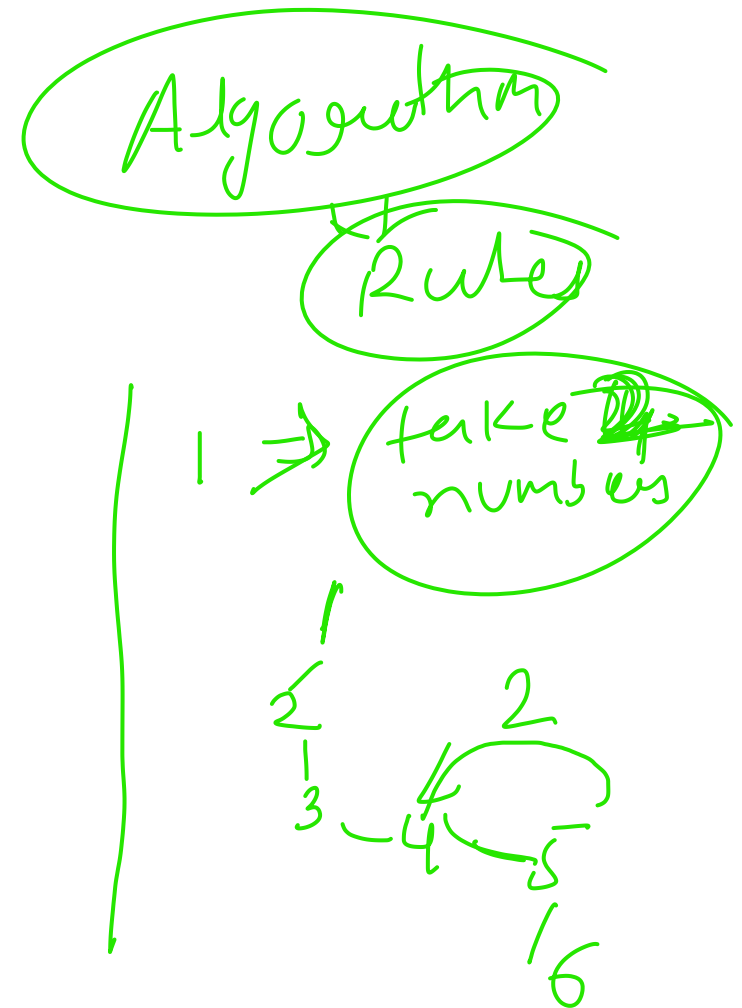
C, C++, Java



Steps to Design an Algorithm

- ✓ Problem Definition
- ✓ Design algorithm
- ✓ Draw flowchart
- ✓ Verification
- Coding(Implementation)
- Analysis(Time and Space Complexity)

algorithm



Algorithm?

→ properties

→ Steps to Design algo

Local

